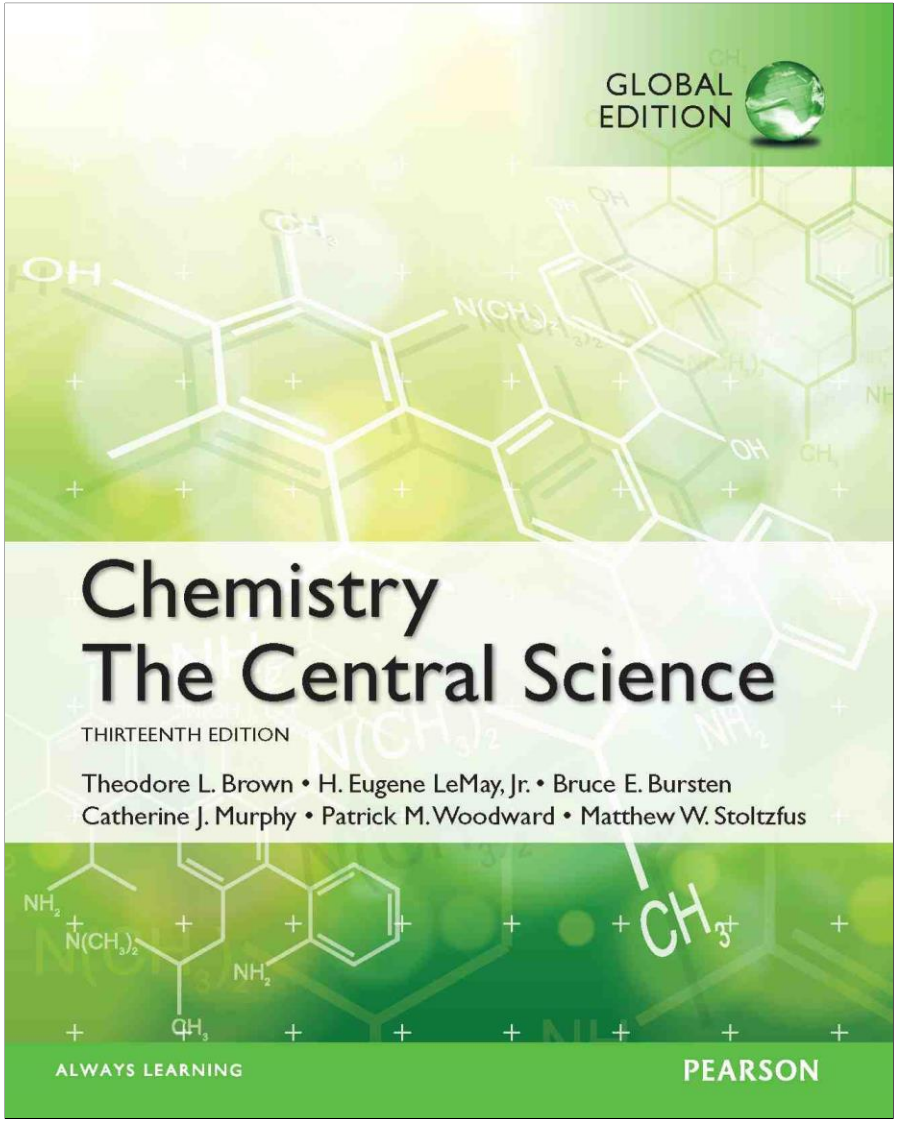

Lecture Presentation

Chapter 15

Chemical Equilibrium

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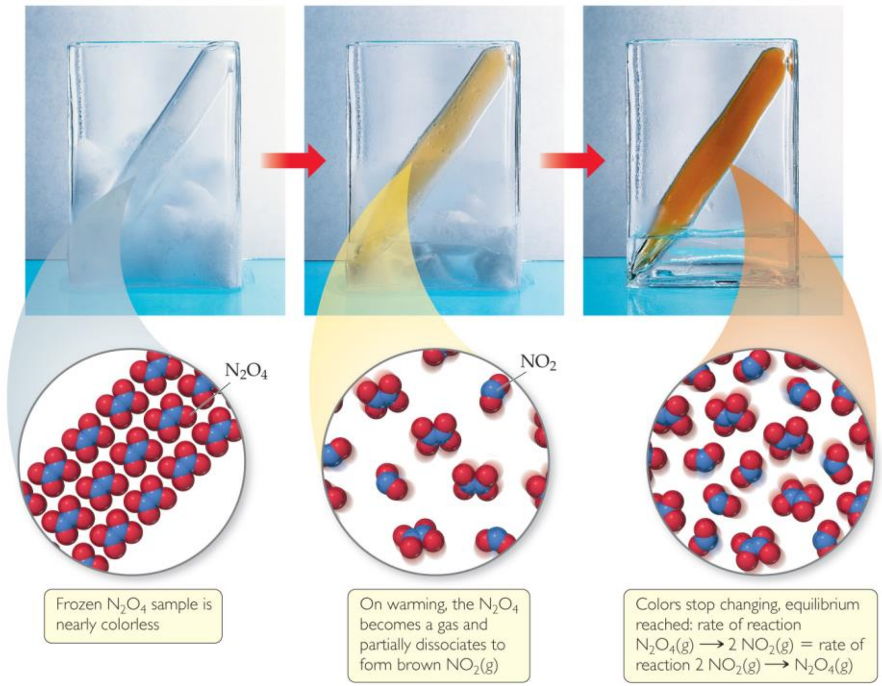
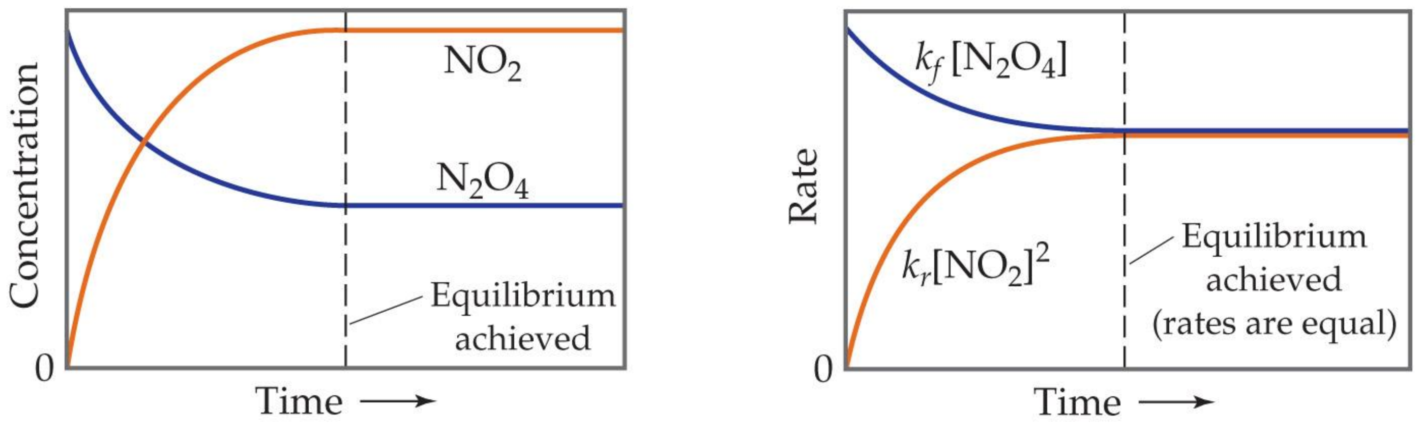
The Concept of Equilibrium



Chemical equilibrium occurs when a reaction and its reverse reaction proceed at the same rate. In the figure above, equilibrium is finally reached in the third picture.

The Concept of Equilibrium



(a) (b)

- As a system approaches equilibrium, both the forward and reverse reactions are occurring.
- At equilibrium, the forward and reverse reactions are proceeding *at the same rate* – **DYNAMIC !**
- Once equilibrium is achieved, the *amount* of each reactant and product remains constant.

Equilibrium

Writing the Equation for an Equilibrium Reaction

Since, in a system at equilibrium, both the forward and reverse reactions are being carried out, we write its equation with a double arrow:

$$N_2O_4(g) \rightleftharpoons 2\ NO_2(g)$$

Comparing Rates

- For the forward reaction
  $N_2O_4(g) \rightarrow 2\ NO_2(g)$
- The rate law is
  Rate = $k_f[N_2O_4]$

- For the reverse reaction
  $2\ NO_2(g) \rightarrow N_2O_4(g)$
- The rate law is
  Rate = $k_r[NO_2]^2$

The Meaning of Equilibrium

- Therefore, at equilibrium

$$\text{Rate}_f = \text{Rate}_r$$

$$k_f[N_2O_4] = k_r[NO_2]^2$$

- Rewriting this, it becomes the expression for the equilibrium constant, K_{eq}.

$$\frac{[NO_2]^2}{[N_2O_4]} = \frac{k_f}{k_r} = \text{a constant}$$

Another Equilibrium—
The Haber Process

- Consider the Haber Process, which is the industrial preparation of ammonia:

  $N_2(g) + 3\ H_2(g) \rightleftharpoons 2\ NH_3(g)$

- The equilibrium constant depends on stoichiometry:

$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3}$$

The Equilibrium Constant

- Consider the generalized reaction

  $a\,\mathrm{A} + b\,\mathrm{B} \rightleftharpoons d\,\mathrm{D} + e\,\mathrm{E}$

- The equilibrium expression for this reaction would be

$$K_c = \frac{[\mathrm{D}]^d[\mathrm{E}]^e}{[\mathrm{A}]^a[\mathrm{B}]^b} \begin{array}{l} \longleftarrow \text{products} \\ \longleftarrow \text{reactants} \end{array}$$

- Also, since pressure is proportional to concentration for gases in a closed system, the equilibrium expression can also be written

$$K_p = \frac{(P_{\mathrm{D}})^d(P_{\mathrm{E}})^e}{(P_{\mathrm{A}})^a(P_{\mathrm{B}})^b}$$

More with Gases and Equilibrium

- We can compare the equilibrium constant based on concentration to the one based on pressure.
- For gases, $PV = nRT$ (the Ideal Gas Law).
- Rearranging, $P = (n/V)RT$; (n/V) is [ ].
- The result is

$$K_p = K_c(RT)^{\Delta n}$$

- where

$$\Delta n = (\text{moles of gaseous product}) - (\text{moles of gaseous reactant})$$

SAMPLE EXERCISE 15.2 Converting between K_c and K_p

For the Haber process,

$$N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$$

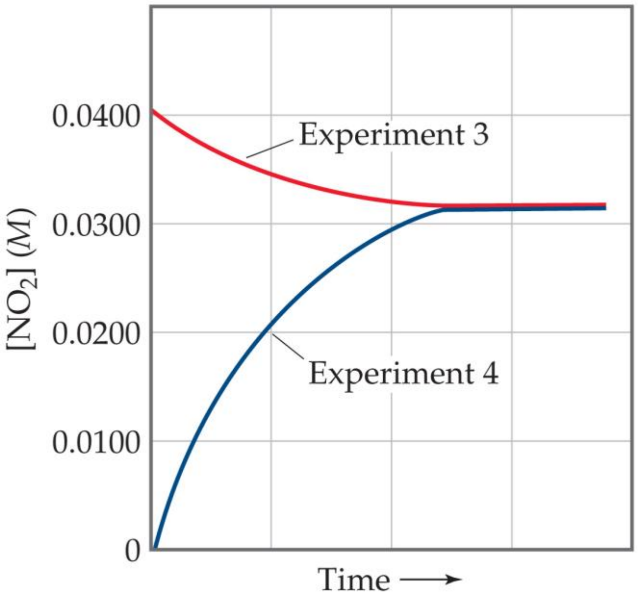
$K_c = 9.60$ at 300 °C. Calculate K_p for this reaction at this temperature.

Equilibrium Can Be Reached from Either Direction

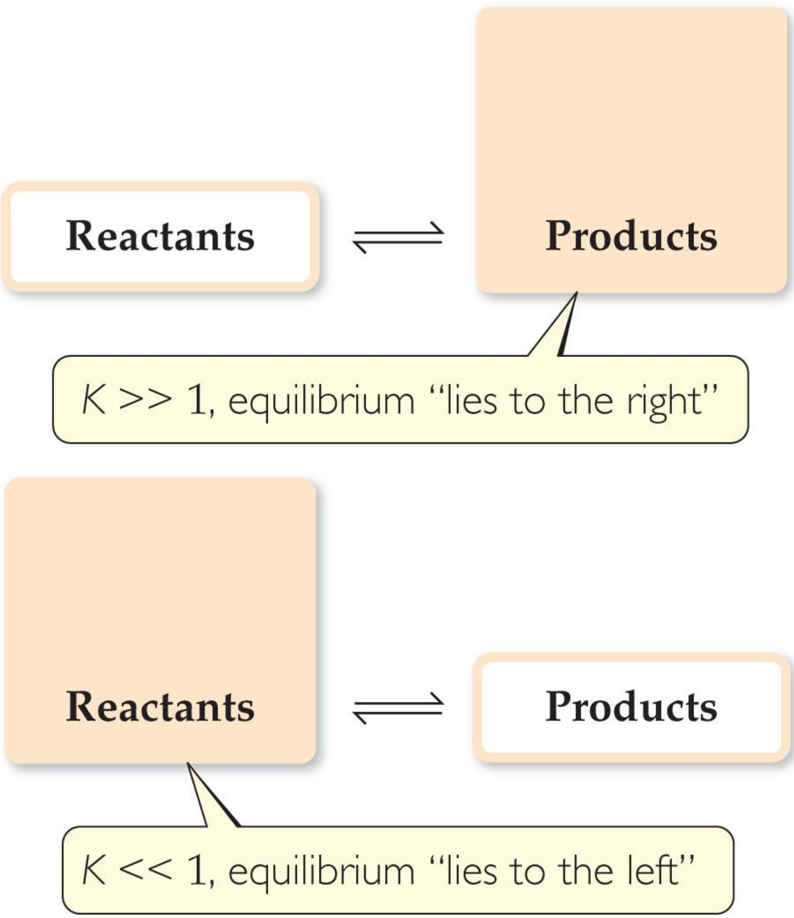
- As you can see, the ratio of $[NO_2]^2$ to $[N_2O_4]$ remains constant at this temperature no matter what the initial concentrations of NO_2 and N_2O_4 are.

Table 15.1 Initial and Equilibrium Concentrations of $N_2O_4(g)$ and $NO_2(g)$ at 100 °C

Experiment	Initial $[N_2O_4]$ (M)	Initial $[NO_2]$ (M)	Equilibrium $[N_2O_4]$ (M)	Equilibrium $[NO_2]$ (M)	K_c
1	0.0	0.0200	0.00140	0.0172	0.211
2	0.0	0.0300	0.00280	0.0243	0.211
3	0.0	0.0400	0.00452	0.0310	0.213
4	0.0200	0.0	0.00452	0.0310	0.213



Magnitude of *K*



- If K>>1, the reaction favors products; products predominate at equilibrium.
- If K<<1, the reaction favors reactants; reactants predominate at equilibrium.

The Direction of the Chemical Equation and *K*

The equilibrium constant of a reaction in the reverse reaction is the reciprocal of the equilibrium constant of the forward reaction:

$$N_2O_4(g) \rightleftharpoons 2\ NO_2(g) \qquad K_c = \frac{[NO_2]^2}{[N_2O_4]} = 0.212 \text{ at } 100\ ^\circ C$$

$$2\ NO_2(g) \rightleftharpoons N_2O_4(g) \qquad K_c = \frac{[N_2O_4]}{[NO_2]^2} = 4.72 \text{ at } 100\ ^\circ C$$

Stoichiometry and Equilibrium Constants

To find the new equilibrium constant of a reaction when the equation has been multiplied by a number, simply raise the original equilibrium constant to that power. Here, the stoichiometry is doubled; the constant is the squared!

$$N_2O_4(g) \rightleftharpoons 2\ NO_2(g) \quad K_c = \frac{[NO_2]^2}{[N_2O_4]} = 0.212 \text{ at } 100\ ^\circ C$$

$$2\ N_2O_4(g) \rightleftharpoons 4\ NO_2(g) \quad K_c = \frac{[NO_2]^4}{[N_2O_4]^2} = (0.212)^2 \text{ at } 100\ ^\circ C$$

$$2\,N_2O_4(g) \rightleftharpoons 4\,NO_2(g) \qquad K_c{}'$$

$$N_2O_4(g) \rightleftharpoons 2\,NO_2(g) \qquad K_c$$

$$N_2O_4(g) \rightleftharpoons 2\,NO_2(g) \qquad K_c$$

$$\boldsymbol{K_c{}' = K_c \times K_c}$$

Consecutive Equilibria

- When two consecutive equilibria occur, the equations can be added to give a single equilibrium.
- The equilibrium constant of the new reaction is the *product* of the two constants:

  $K_3 = K_1 \times K_2$

- Example

➢ $2\ NOBr \rightleftharpoons 2\ NO + Br_2$ $\quad K_1 = 0.014$

➢ $Br_2 + Cl_2 \rightleftharpoons 2\ BrCl$ $\quad K_2 = 7.2$

➢ $2\ NOBr + Cl_2 \rightleftharpoons 2\ NO + 2\ BrCl$

$K_3 = K_1 \times K_2 = 0.014 \times 7.2 = 0.10$

SAMPLE EXERCISE 15.1 Writing Equilibrium-Constant Expressions

Write the equilibrium expression for K_c for the following reactions:

(a) $2\,O_3(g) \rightleftharpoons 3\,O_2(g)$

(b) $2\,NO(g) + Cl_2(g) \rightleftharpoons 2\,NOCl(g)$

(c) $Ag^{+}(aq) + 2\,NH_3(aq) \rightleftharpoons Ag(NH_3)_2^{+}(aq)$

SAMPLE EXERCISE 15.4 Combining Equilibrium Expressions

Given the reactions

$$HF(aq) \rightleftharpoons H^+(aq) + F^-(aq) \qquad K_c = 6.8 \times 10^{-4}$$

$$H_2C_2O_4(aq) \rightleftharpoons 2\,H^+(aq) + C_2O_4^{2-}(aq) \qquad K_c = 3.8 \times 10^{-6}$$

determine the value of K_c for the reaction

$$2\,HF(aq) + C_2O_4^{2-}(aq) \rightleftharpoons 2\,F^-(aq) + H_2C_2O_4(aq)$$

Homogeneous vs. Heterogeneous

- **Homogeneous equilibria** occur when all reactants and products are in the same phase.
- **Heterogeneous equilibria** occur when something in the equilibrium is in a different phase.
- The value used for the concentration of a pure substance is always 1.

The Decomposition of $CaCO_3$—A Heterogeneous Equilibrium

- The equation for the reaction is

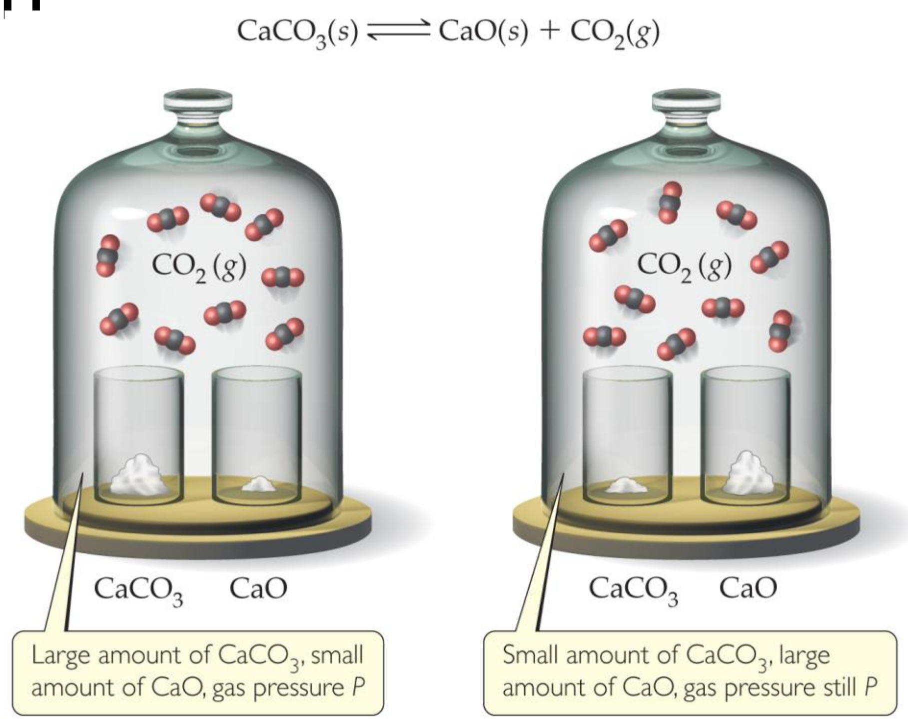
  $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

- This results in

  $K_c = [CO_2]$

  and

  $K_p = P_{CO_2}$



SAMPLE EXERCISE 15.5 Writing Equilibrium-Constant Expressions for Heterogeneous Reactions

Write the equilibrium-constant expression K_c for

(a) $CO_2(g) + H_2(g) \rightleftharpoons CO(g) + H_2O(l)$

(b) $SnO_2(s) + 2\,CO(g) \rightleftharpoons Sn(s) + 2\,CO_2(g)$

Deducing Equilibrium Concentrations

ICE approach

I - Initial Concentration(s)

C - Change in Concentration(s) when equilibrium is reached in comparison to the initial concentration(s)

E – Concentrations at Equilibrium

An Example

A closed system initially containing 1.000×10^{-3} M H_2 and 2.000×10^{-3} M I_2 at 448ºC is allowed to reach equilibrium. Analysis of the equilibrium mixture shows that the concentration of HI is 1.87×10^{-3} M. Calculate K_c at 448ºC for the reaction taking place, which is

$$H_2(g) + I_2(g) \rightleftharpoons 2\ HI(g)$$

What Do We Know?

	$[H_2]$, *M*	$[I_2]$, *M*	[HI], *M*
Initial Concentration	1.000×10^{-3}	2.000×10^{-3}	0
Change in Concentration	**- (1/2)** 1.87×10^{-3}	**- (1/2)** 1.87×10^{-3}	**+** 1.87×10^{-3}
Concentration at Equilibrium	$(1.000 - 0.935) \times 10^{-3}$ $= \mathbf{0.065 \times 10^{-3}}$	$(2.000 - 0.935) \times 10^{-3}$ $= \mathbf{1.065 \times 10^{-3}}$	$\mathbf{1.87 \times 10^{-3}}$

And, therefore, the equilibrium constant…

$$K_c = \frac{[HI]^2}{[H_2]\,[I_2]}$$

$$= \frac{(1.87 \times 10^{-3})^2}{(6.5 \times 10^{-5})(1.065 \times 10^{-3})}$$

$$= 51$$

Calculating Equilibrium Concentrations
(using the same ICE approach)

If you know the equilibrium constant, you can find equilibrium concentrations from initial concentrations and changes (based on stoichiometry).

A 1.000 L flask is filled with 1.000 mol of $H_2(g)$ and 2.000 mol of $I_2(g)$ at 448 °C. Given a K_c of 50.5 at 448 °C, what are the equilibrium concentrations of H_2, I_2, and HI?

$$H_2(g) + I_2(g) \rightleftharpoons 2\ HI(g)$$

initial concentration (M)	**1.000**	**2.000**	**0**
change in concentration (M)	**$-x$**	**$-x$**	**$+2x$**
equilibrium concentration (M)	**$1.000 - x$**	**$2.000 - x$**	**$2x$**

- Set up the equilibrium constant expression, filling in equilibrium concentrations from the table.

$$K_c = \frac{[HI]^2}{[H_2][I_2]} = \frac{(2x)^2}{(1.000 - x)(2.000 - x)} = 50.5$$

- Solving for *x* is done using the quadratic formula, resulting in *x* = 2.323 *or* 0.935.

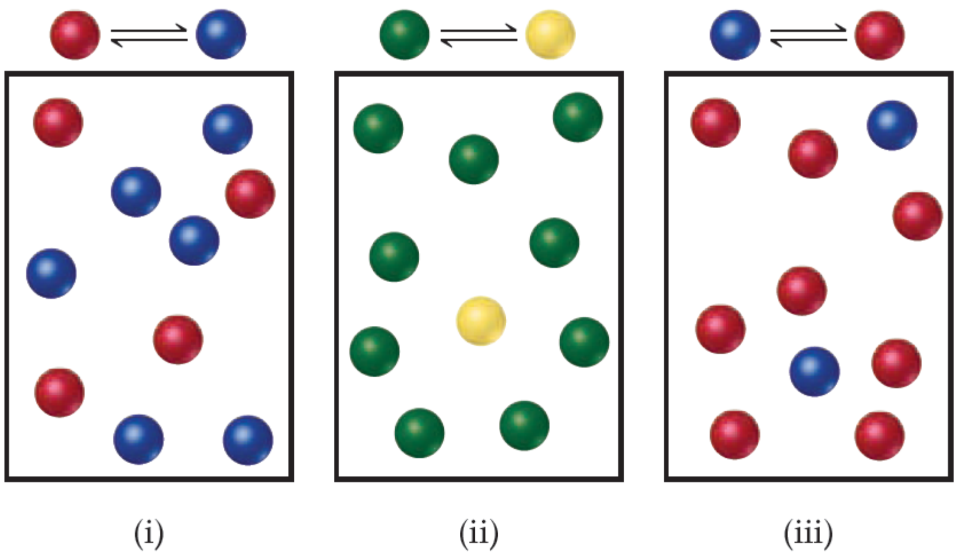
Since *x* must be subtracted from 1.000 *M*, 2.323 makes no physical sense. (It results in a negative concentration!) The value *must* be 0.935.

So

- $[H_2]_{eq}$ = 1.000 – 0.935 = 0.065 M
- $[I_2]_{eq}$ = 2.000 – 0.935 = 1.065 M
- $[HI]_{eq}$ = 2(0.935) = 1.87 M

SAMPLE EXERCISE 15.3 Interpreting the Magnitude of an Equilibrium Constant

The following diagrams represent three systems at equilibrium, all in the same-size containers. **(a)** Without doing any calculations, rank the systems in order of increasing K_c. **(b)** If the volume of the containers is 1.0 L and each sphere represents 0.10 mol, calculate K_c for each system.

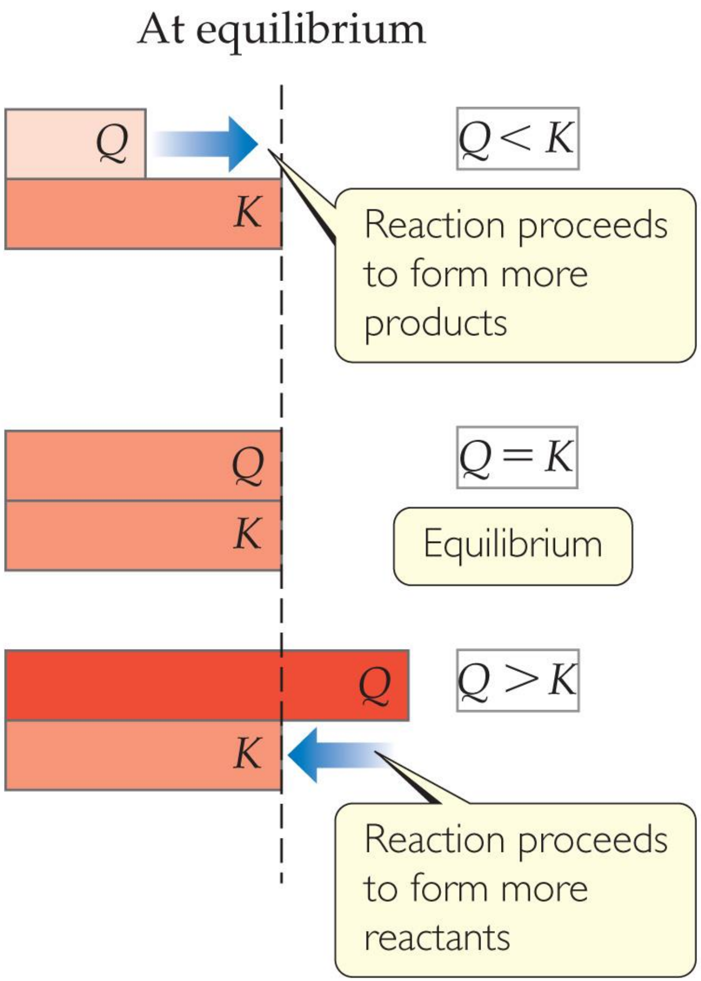


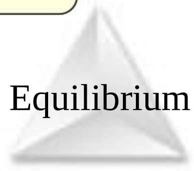
Is a Mixture in Equilibrium? Which Way Does the Reaction Go?

- To answer these questions, we calculate the **reaction quotient**, *Q*.
- Q looks like the equilibrium constant, *K*, but the values used to calculate it are the current conditions, not necessarily those for equilibrium.
- To calculate *Q*, one substitutes the initial concentrations of reactants and products into the equilibrium expression.

Comparing *Q* and *K*

- Nature wants $Q = K$.
- If $Q < K$, nature will make the reaction proceed to products.
- If $Q = K$, the reaction is in equilibrium.
- If $Q > K$, nature will make the reaction proceed to reactants.





Exercise Predicting the Direction of Approach to Equilibrium

At 448 °C the equilibrium constant K_c for the reaction

$$H_2(g) + I_2(g) \rightleftharpoons 2\,HI(g)$$

is 50.5. Predict in which direction the reaction proceeds to reach equilibrium if we start with 2.0×10^{-2} mol of HI, 1.0×10^{-2} mol of H_2, and 3.0×10^{-2} of I_2 in a 2.00-L container.

Exercise Predicting the Direction of Approach to Equilibrium

At 448 °C the equilibrium constant K_c for the reaction

$$H_2(g) + I_2(g) \rightleftharpoons 2\,HI(g)$$

is 50.5. Predict in which direction the reaction proceeds to reach equilibrium if we start with 2.0×10^{-2} mol of HI, 1.0×10^{-2} mol of H_2, and 3.0×10^{-2} of I_2 in a 2.00-L container.

Solution

The current concentrations are

$$[HI] = 2.0 \times 10^{-2}\ \text{mol}/2.00\ \text{L} = 1.0 \times 10^{-2}\ M$$
$$[H_2] = 1.0 \times 10^{-2}\ \text{mol}/2.00\ \text{L} = 5.0 \times 10^{-3}\ M$$
$$[I_2] = 3.0 \times 10^{-2}\ \text{mol}/2.00\ \text{L} = 1.5 \times 10^{-2}\ M$$

The reaction quotient is therefore

$$Q_c = \frac{[HI]^2}{[H_2][I_2]} = \frac{(1.0 \times 10^{-2})^2}{(5.0 \times 10^{-3})(1.5 \times 10^{-2})} = 1.3$$

Because $Q_c < K_c$, the concentration of HI must increase and the concentrations of H_2 and I_2 must decrease to reach equilibrium; the reaction as written proceeds left to right to attain equilibrium.

Le Châtelier's Principle

Henri-Louis Le Châtelier (1850–1936)

Le Châtelier's Principle

"If a system at equilibrium is disturbed by a change in temperature, pressure (volume), or the concentration of one of the components, the system will shift its equilibrium position so as to counteract (offset) the effect of the disturbance."

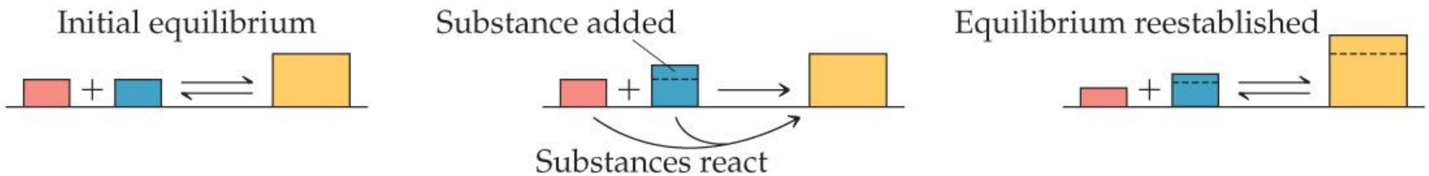
Using LeChâtelier's Principle qualitatively to predict shifts in equilibrium based on changes in conditions.

Le Châtelier's Principle

If a system at equilibrium is disturbed by a change in **concentration, pressure,** or **temperature,** the system will shift its equilibrium position so as to counter the effect of the disturbance.

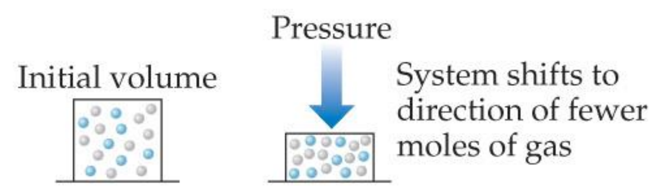
Concentration: adding or removing a reactant or product

If a substance is added to a system at equilibrium, the system reacts to consume some of the substance. If a substance is removed from a system, the system reacts to produce more of substance.



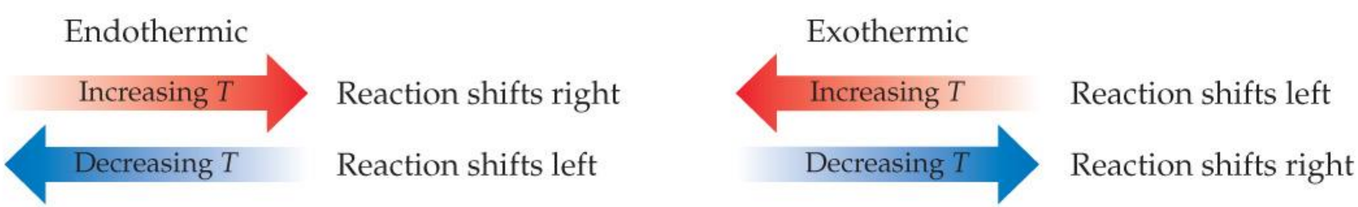
Pressure: changing the pressure by changing the volume

At constant temperature, reducing the volume of a gaseous equilibrium mixture causes the system to shift in the direction that reduces the number of moles of gas.



Temperature:

If the temperature of a system at equilibrium is increased, the system reacts as if we added a reactant to an endothermic reaction or a product to an exothermic reaction. The equilibrium shifts in the direction that consumes the "excess reactant," namely heat.

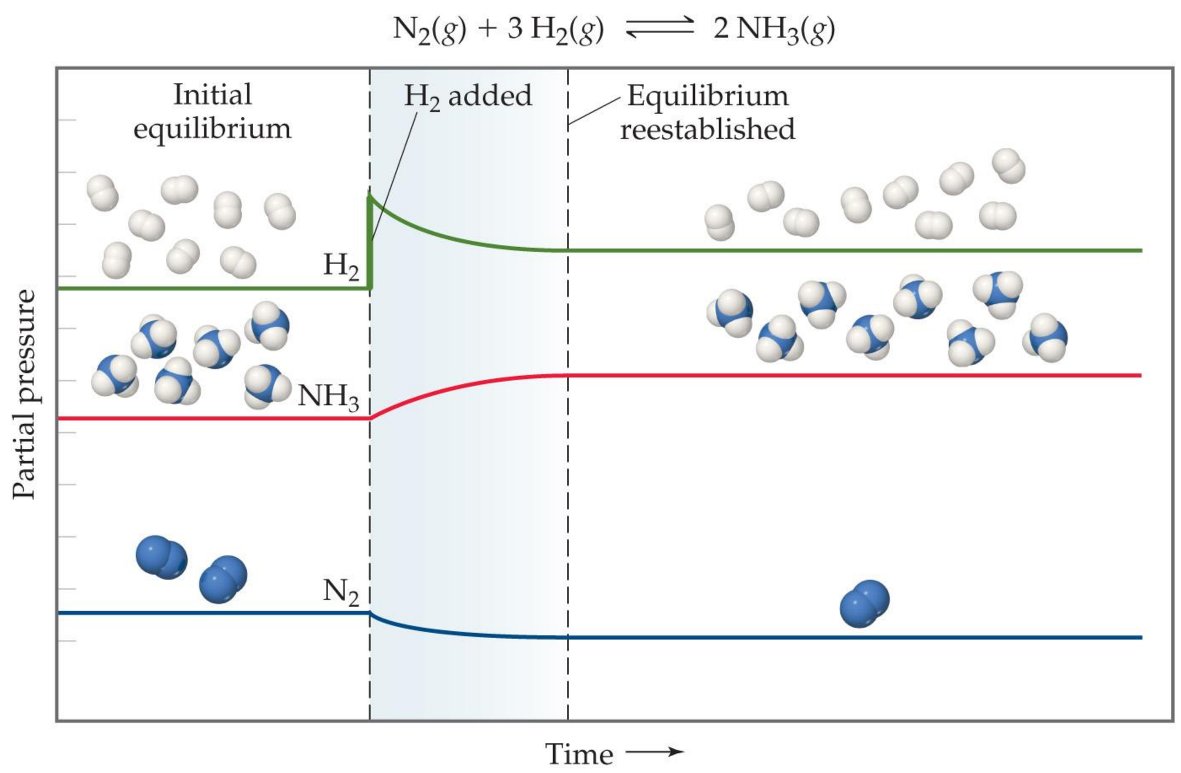


Change in Reactant or Product Concentration

- If the system is in equilibrium

adding a reaction component will result in some of it being used up.

removing a reaction component will result in some if it being produced.



Equilibrium

Using LeChâtelier's Principle qualitatively to predict shifts in equilibrium based on changes in conditions.

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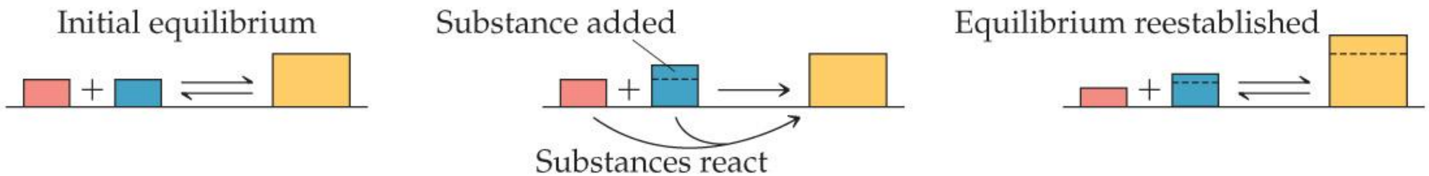

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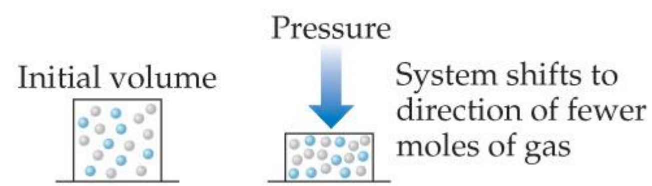

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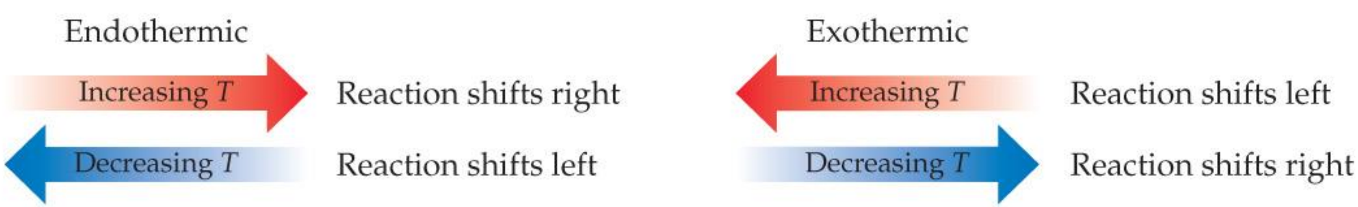
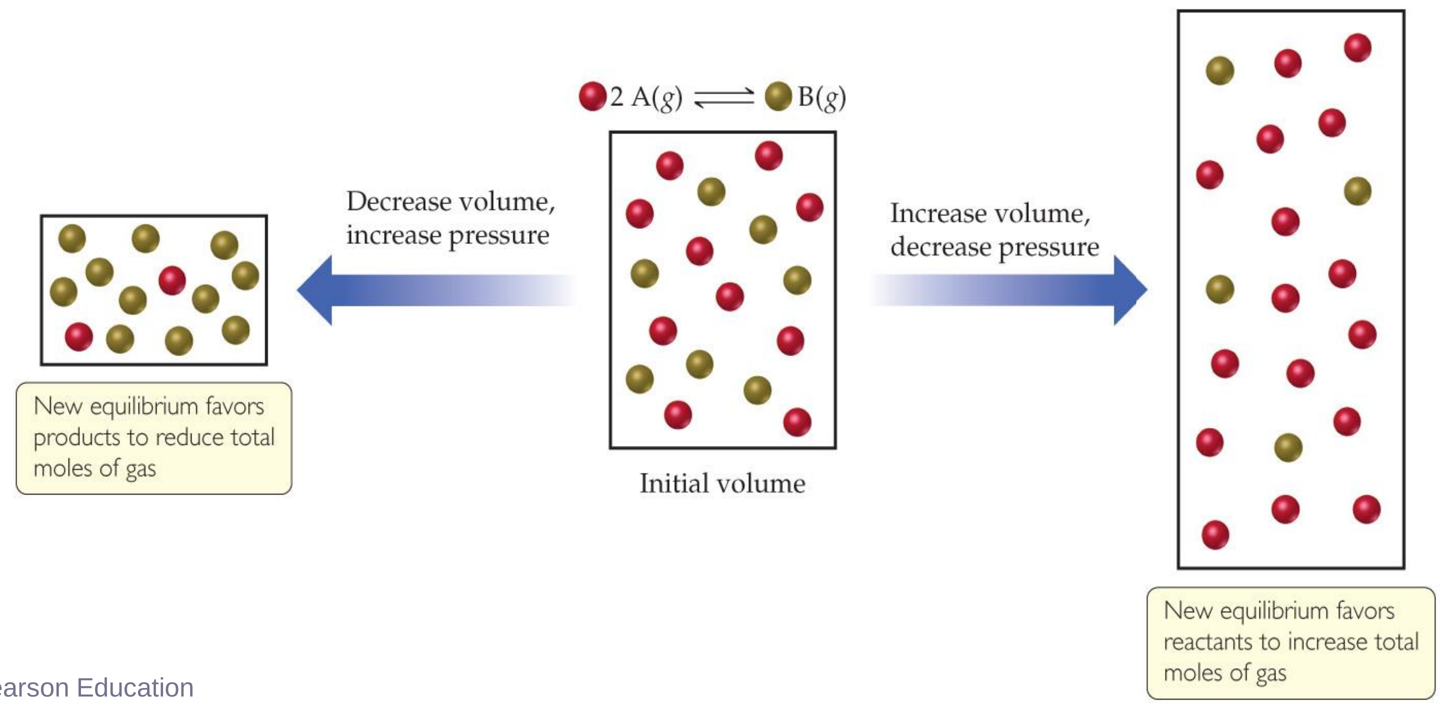


Change in Volume or Pressure

- When gases are involved in an equilibrium, a change in pressure or volume will affect equilibrium:
  – Higher volume or lower pressure favors the side of the equation with more moles (and vice-versa).



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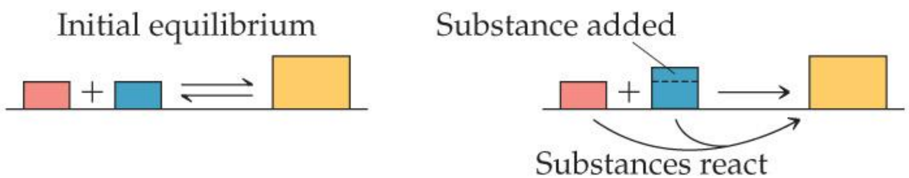

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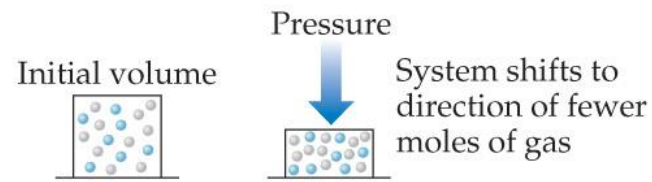


Temperature:

If the temperature of a system at equilibrium is increased, the system reacts as if we added a reactant to an endothermic reaction or a product to an exothermic reaction. The equilibrium shifts in the direction that consumes the "excess reactant," namely heat.

Change in Temperature

- Is the reaction endothermic or exothermic as written? That matters!
- **Exothermic**: Heat acts *like* a product; adding heat drives a reaction toward reactants.

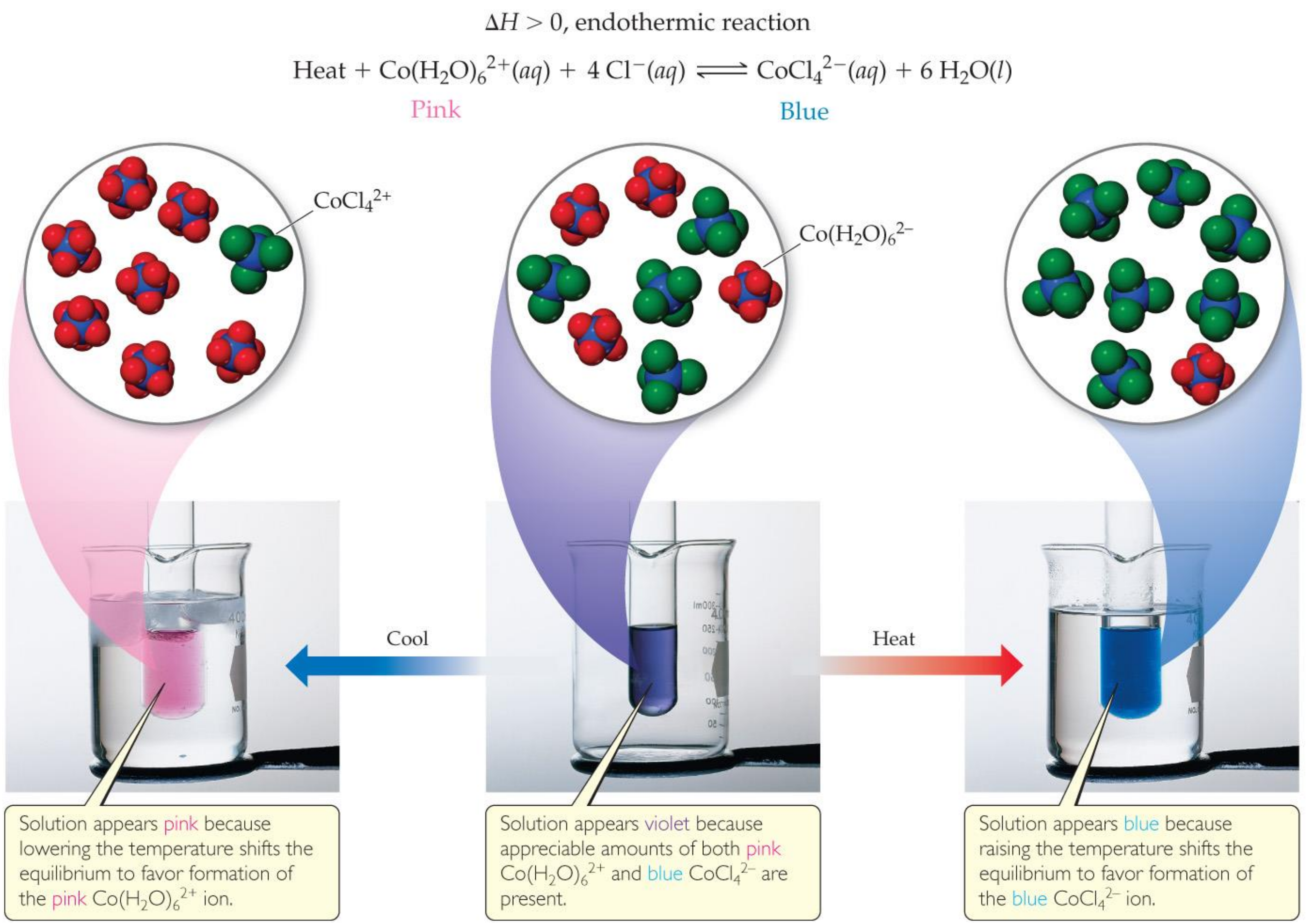
Reactants → Products + heat

- Endothermic: Heats acts *like* a reactant; adding heat drives a reaction toward products.

Reactants + heat → Products

An Endothermic Equilibrium

$$\underset{\text{Pale pink}}{Co(H_2O)_6^{2+}(aq)} + 4\,Cl^-(aq) \rightleftharpoons \underset{\text{Deep blue}}{CoCl_4^{2-}(aq)} + 6\,H_2O(l) \qquad \Delta H > 0$$



Equilibrium

An Exothermic Equilibrium

$$N_2(g) + 3\ H_2(g) \longrightarrow 2\ NH_3(g) \qquad \Delta H = -92\ \text{kJ/mol}$$

- The Haber Process for producing ammonia from the elements is exothermic.
- One would think that cooling down the reactants would result in more product (larger K_{eq}) but slower reaction (smaller k).
- The activation energy for this reaction is high!
- This is the *one* instance where a system in equilibrium can be affected by a catalyst!

Changes in Temperature

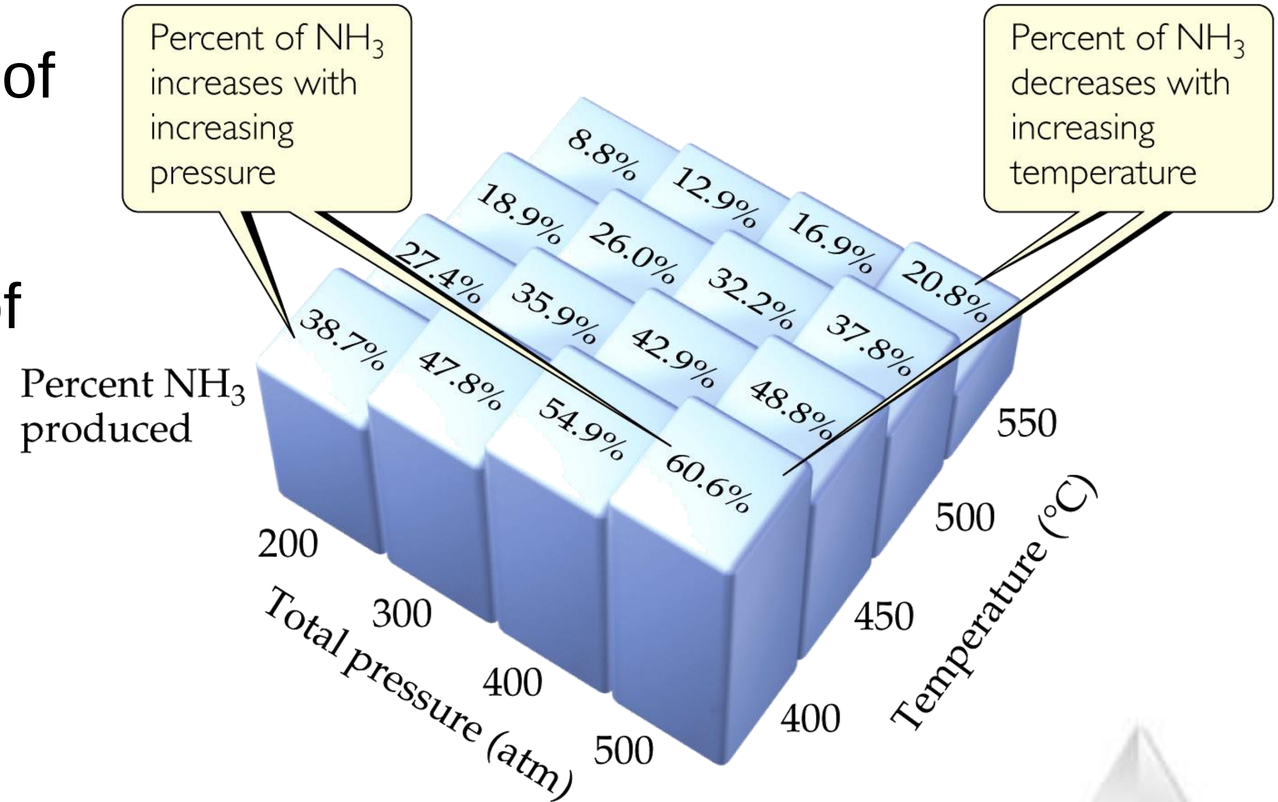
$$N_2(g) + 3\ H_2(g) \longrightarrow 2\ NH_3(g) \qquad \Delta H = -92\ \text{kJ/mol}$$

Temperature (°C)	K_p
300	4.34×10^{-3}
400	1.64×10^{-4}
450	4.51×10^{-5}
500	1.45×10^{-5}
550	5.38×10^{-6}
600	2.25×10^{-6}

The Haber Process

$$N_2(g) + 3\,H_2(g) \longrightarrow 2\,NH_3(g) \qquad \Delta H = -92\ \text{kJ/mol}$$

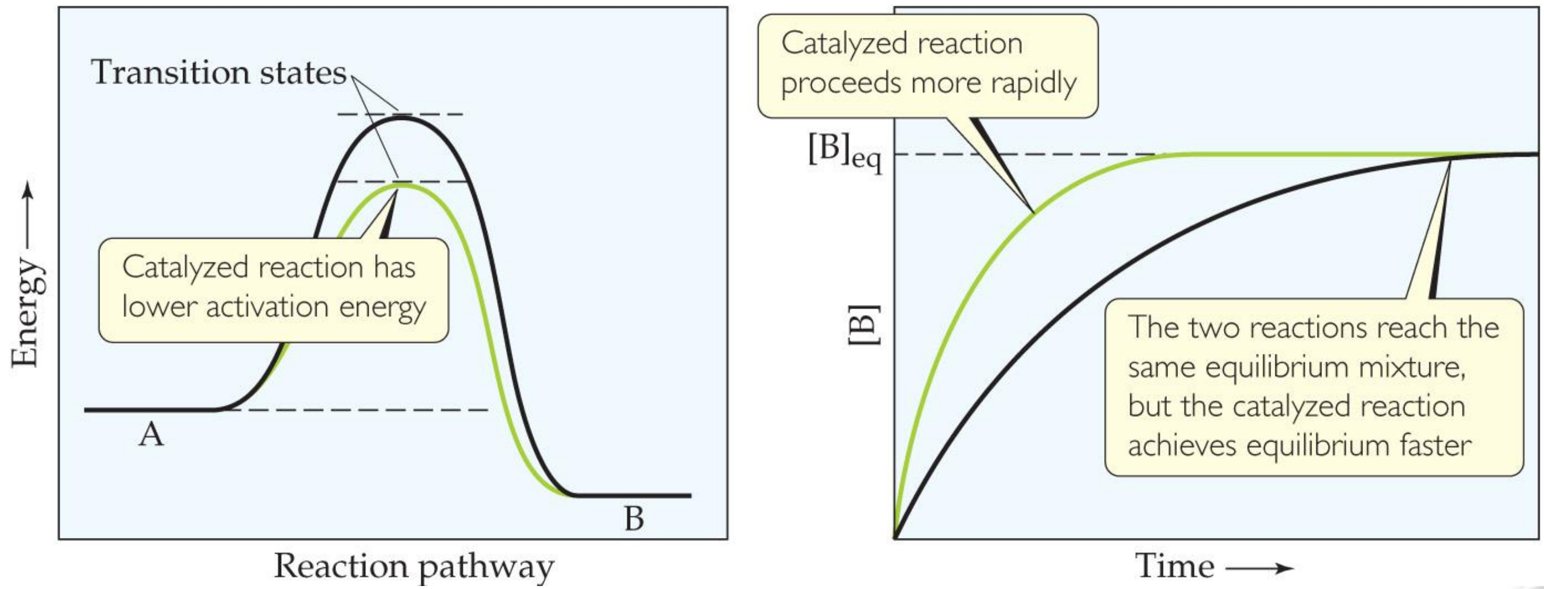
The transformation of nitrogen and hydrogen into ammonia (NH_3) is of tremendous significance in agriculture, where ammonia-based fertilizers are of utmost importance.



Equilibrium

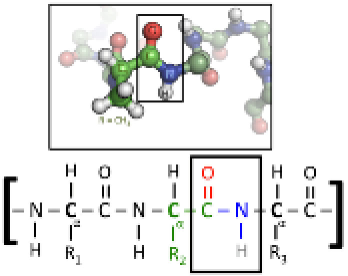
Catalysts

- **Catalysts** increase the rate of both the forward *and* reverse reactions.
- Equilibrium is achieved faster, but the equilibrium composition remains unaltered.
- Activation energy is lowered, allowing equilibrium to be established at lower temperatures.



uilibrium

N_2



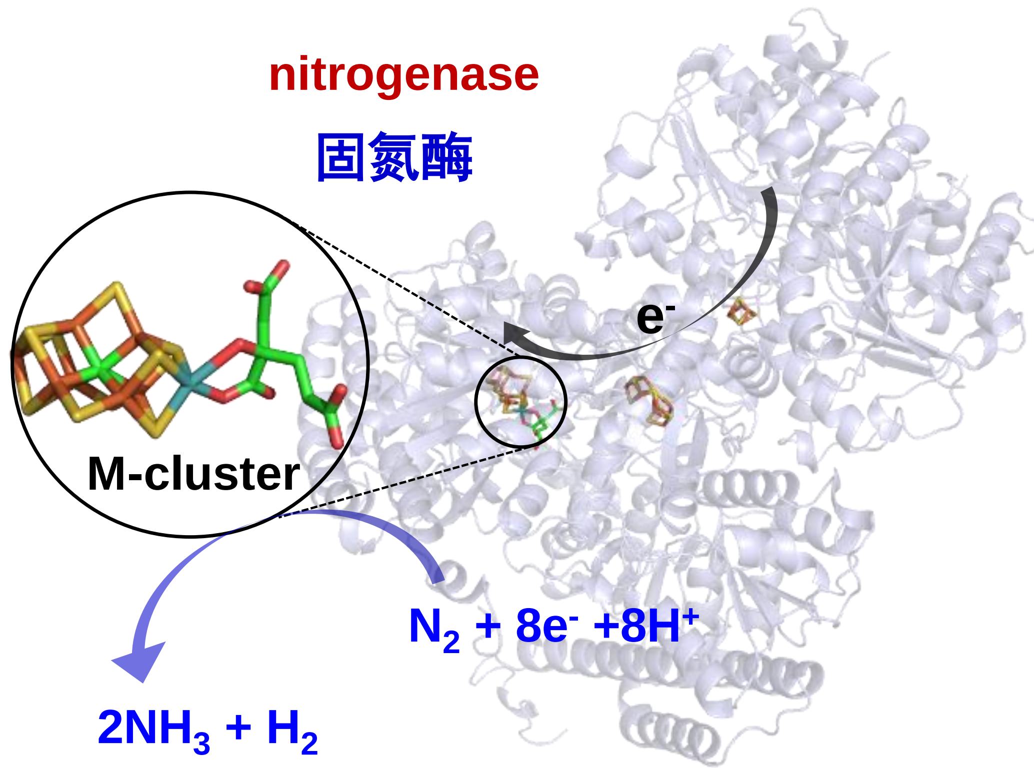
Protein 蛋白质

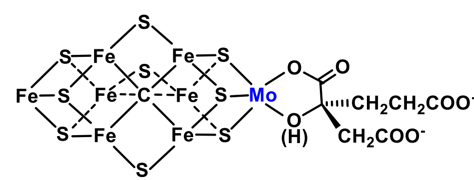
空气中含有
78%氮气N_2

78% N_2 in air

:N:::N:
:N≡N:

根瘤菌 (Rhizobia)





Fe_7MoS_9C

Equilibrium

N_2

一亩大豆固氮约 8kg/year

大豆玉米复合种植
bean and corn

人工合成氨 **Haber–Bosch process**

$$N_2 + 3\,H_2 \longrightarrow 2\,NH_3 \quad \Delta H^\circ = -91.8\ \mathrm{kJ/mol}$$

230 million tonnes per year

Equilibrium

Haber–Bosch process

Nobel Prize in chemistry (1918)

Fritz Haber (1868-1934)

Nobel Prize in chemistry (1931)

Carl Bosch (1874-1940)

Nobel Prize in chemistry (2007)

Gerhard Ertl (1936-)

Le Châtelier's Principle

System	Change	Result
$CO_2 + H_2 \rightarrow H_2O(g) + CO$	A drying agent is added to absorb H_2O	**Shift to the right**. Continuous removal of a product will force any reaction to the right
$H_2(g) + I_2(g) \rightarrow 2HI(g)$	Some nitrogen gas is added	**No change**; N_2 is not a component of this reaction system.
$NaCl(s) + H_2SO_4(l) \rightarrow Na_2SO_4(s) + HCl(g)$	Reaction is carried out in an open container	Because HCl is a gas that can escape from the system, the reaction is forced to the **right**. This is the basis for the commercial production of hydrochloric acid.
$H_2O(l) \rightarrow H_2O(g)$	Water evaporates from an open container	Continuous removal of water vapor forces the reaction to the **right**, so equilibrium is never achieved.
$HCN(aq) \rightarrow H^+(aq) + CN^-(aq)$	The solution is diluted	Shift to **right**; the product $[H^+][CN^-]$ diminishes more rapidly than does [HCN].
$AgCl(s) \rightarrow Ag^+(aq) + Cl^-(aq)$	Some NaCl is added to the solution	Shift to **left** due to increase in Cl^- concentration. This is known as the common ion effect on solubility.
$N_2 + 3\ H_2 \rightarrow 2\ NH_3$	A catalyst is added to speed up this reaction	**No change**. Catalysts affect only the rate of a reaction; they have no effect at all on the composition of the equilibrium state.

Consider the equilibrium

$$N_2O_4(g) \rightleftharpoons 2\,NO_2(g) \qquad \Delta H^\circ = 58.0\text{ kJ}$$

In which direction will the equilibrium shift when **(a)** N_2O_4 is added, **(b)** NO_2 is removed, **(c)** the pressure is increased by addition of $N_2(g)$, **(d)** the volume is increased, **(e)** the temperature is decreased?

GIVE IT SOME THOUGHT

Use Le Châtelier's principle to explain why the equilibrium vapor pressure of a liquid increases with increasing temperature.

A. Evaporation is an exothermic process and an increasing temperature shifts the equilibrium $H_2O(l) \rightleftarrows H_2O(g)$ to the right and increases the equilibrium vapor pressure.

$\boxed{\text{B.}}$ Evaporation is an endothermic process and an increasing temperature shifts the equilibrium $H_2O(l) \rightleftarrows H_2O(g)$ to the right and increases the equilibrium vapor pressure.

C. Evaporation is an exothermic process and an increasing temperature shifts the equilibrium $H_2O(l) \rightleftarrows H_2O(g)$ to the left and increases the equilibrium vapor pressure.

D. Evaporation is an endothermic process and an increasing temperature shifts the equilibrium $H_2O(l) \rightleftarrows H_2O(g)$ to the left and increases the equilibrium vapor pressure.

At equilibrium, the rate of the forward reaction is ________ the rate of the reverse reaction.

a. equal to
b. slower than
c. faster than
d. the reverse of

Equilibrium

The reaction quotient Q is usually represented by

a. [reactants] / [products].
b. [products] / [reactants].
c. [reactants] × [products].
d. [reactants] + [products].

Equilibrium

If the value of the equilibrium constant is large, then ________ will mostly be present at equilibrium.

a. reactants
b. products
c. catalysts
d. shrapnel

If the value of the equilibrium constant is small, then ________ will mostly be present at equilibrium.

a. reactants
b. products
c. catalysts
d. shrapnel

Q = the reaction quotient
K = the equilibrium constant
At equilibrium, which is true?

a. $Q > K$
b. $Q < K$
c. $Q = K$
d. $Q^2 = K$

Equilibrium

Equilibrium constants typically have units of

a. M.
b. M^2.
c. M^{1-}.
d. None of the above

Reaction quotients for heterogeneous equilibria do not include concentrations of

a. pure liquids.
b. pure solids.
c. Both of the above
d. Neither of the above

Equilibrium

$HA \leftrightarrow H^+ + A^-$

$[HA] = 1.65 \times 10^{-2}$ M and $[H^+] = [A^-] = 5.44 \times 10^{-4}$ M at equilibrium. K_c = __________.

a. 1.79×10^{-2}

b. 1.79×10^{-3}

c. 1.79×10^{-4}

$\boxed{\text{d.}}$ 1.79×10^{-5}

Equilibrium

$K_P = K_C$ when

a. the reaction is at equilibrium.
b. the reaction is exothermic.
c. all of the gases present are at the same pressure.
d. the number of moles of gas on both sides of the balanced equation is the same.

$CO_2 + H_2 \leftarrow\rightarrow CO + H_2O$

If all species are gases and H_2 is added, the concentration of CO at equilibrium will

a. increase.
b. decrease.
c. remain unchanged.
d. disappear.

Equilibrium

$CO_2 + H_2 \leftrightarrow CO + H_2O$

If all species are gases and H_2O is added, the equilibrium concentration of CO

a. will increase.

$\boxed{\text{b.}}$ will decrease.

c. will remain unchanged.

d. will disappear.

Equilibrium

$CO_2 + H_2 \leftrightarrow CO + H_2O$

If all species are gases and CO_2 is removed, the [CO] at equilibrium will

a. increase.

$\boxed{\text{b.}}$ decrease.

c. remain unchanged.

d. disappear.

Equilibrium

$CO_2 + H_2 \leftrightarrow CO + H_2O$

Increasing the temperature of this endothermic reaction will ________ [CO] at equilibrium.

a. increase
b. decrease
c. not change
d. eradicate

Equilibrium

$CO_2 + H_2 \leftrightarrow CO + H_2O$

If all species are gases and the container is compressed, the amount of CO will

a. increase.

b. decrease.

$\boxed{\text{c.}}$ remain unchanged.

d. vanish.

Equilibrium

$CO_2 + H_2 \leftrightarrow CO + H_2O$

Adding a catalyst to this reaction will cause the [CO] at equilibrium to

a. increase.

b. decrease.

c. remain unchanged.

d. cease to exist.

Equilibrium